Accelerate Test Execution with Adaptive Test

Reference: RA_453

Unlock Smarter Testing with DAS

Welcome to the Adaptive Test DAS Demo — a powerful illustration of how test programs can be made smarter, more flexible, and responsive to real-time conditions. Whether you're running it locally or envisioning a scaled deployment on remote servers, this demo is built to showcase adaptability.

What This Demo Demonstrates

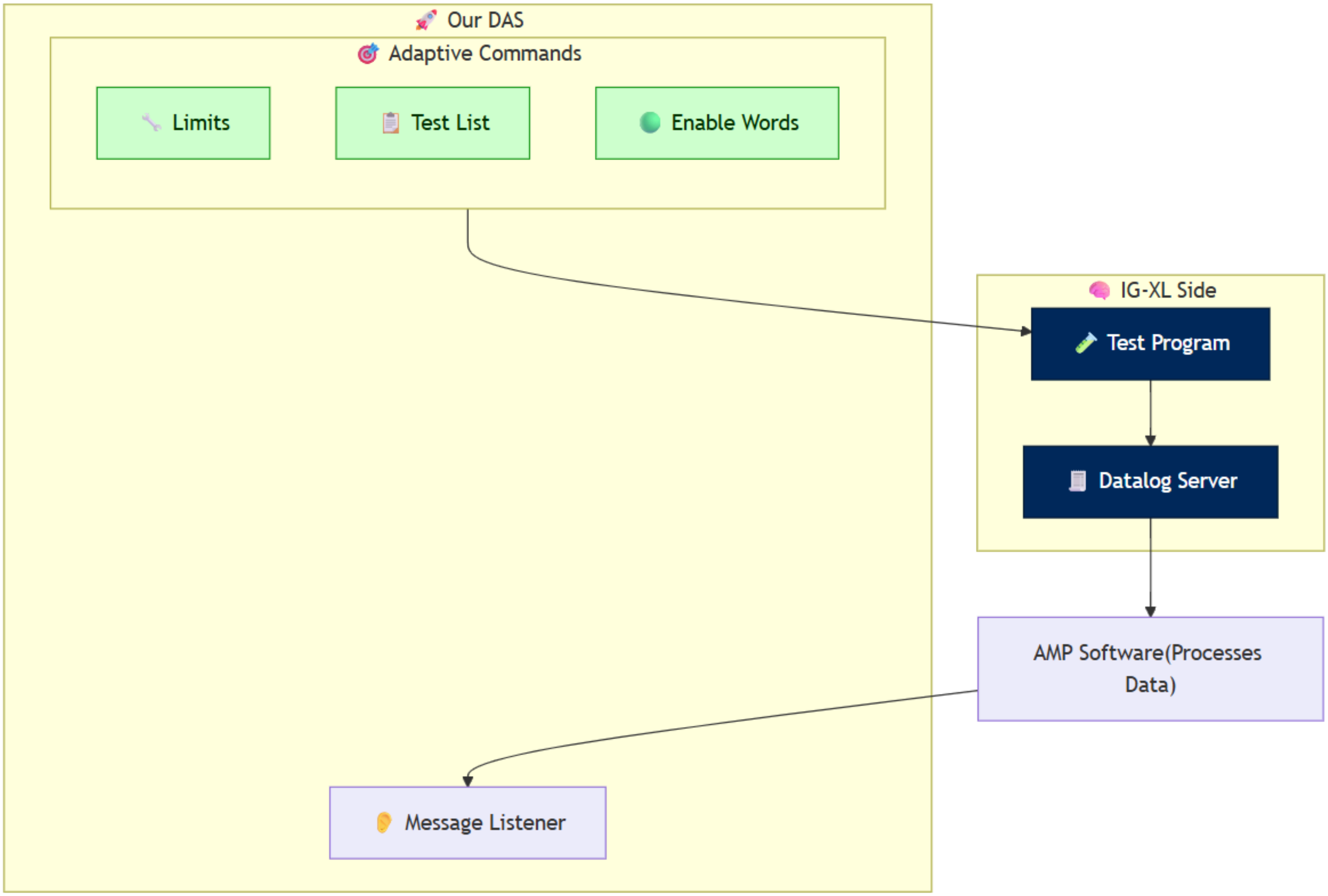
This demo highlights three key capabilities of our Data Analytics Solution (DAS):

1. **Enabling or disabling specific tests** — streamline test execution based on context or priorities
2. **Modifying test limits dynamically** — improve accuracy and adapt to production variability
3. **Controlling enable words** — activate or deactivate sub-flows conditionally

Additionally, the system offers robust observability:

- Monitor messages sent by the tester in real time
- Understand communication patterns driven by adaptive logic

Software Overview

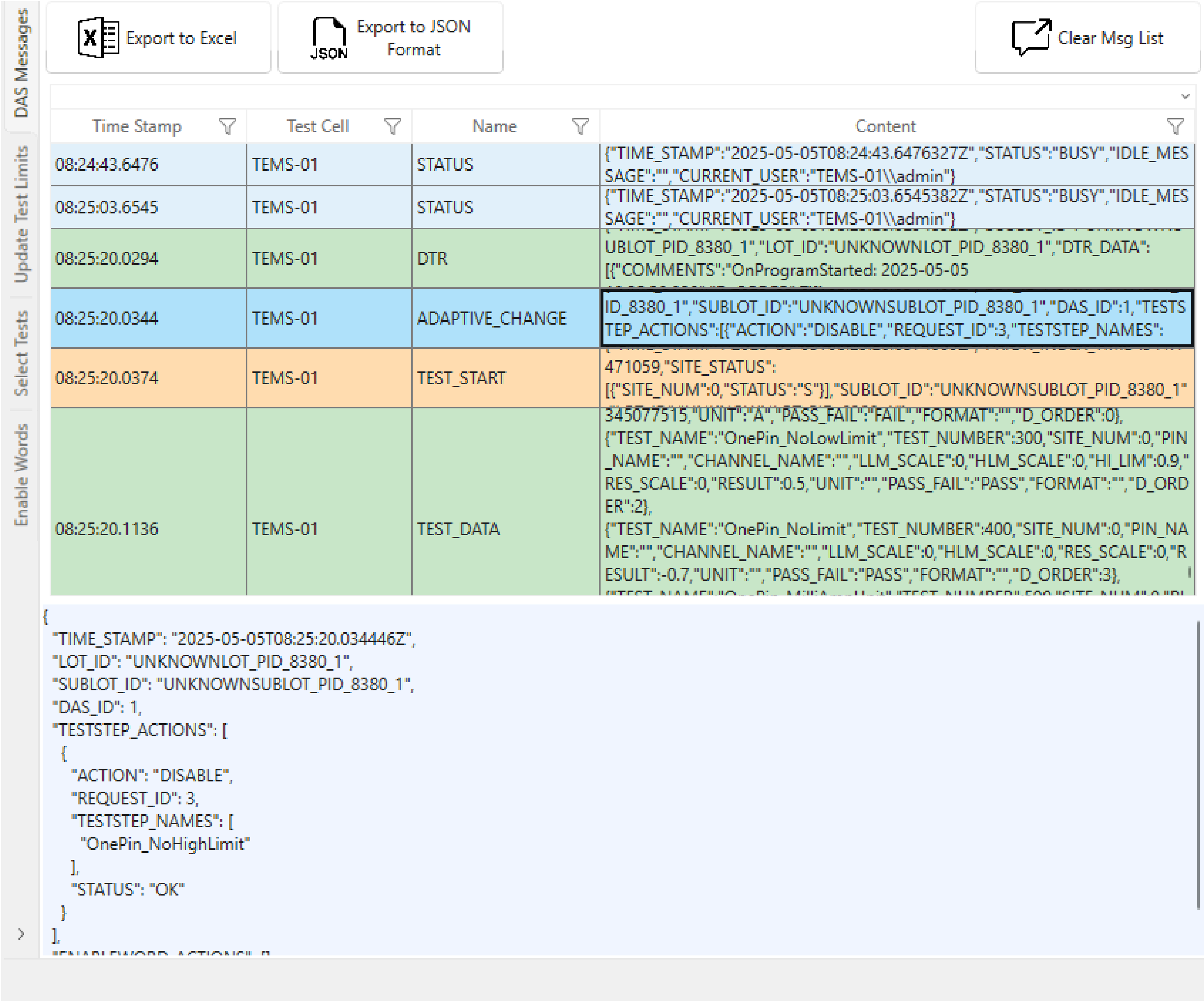


Explore the Demo Software

The demo application includes four dedicated sections:

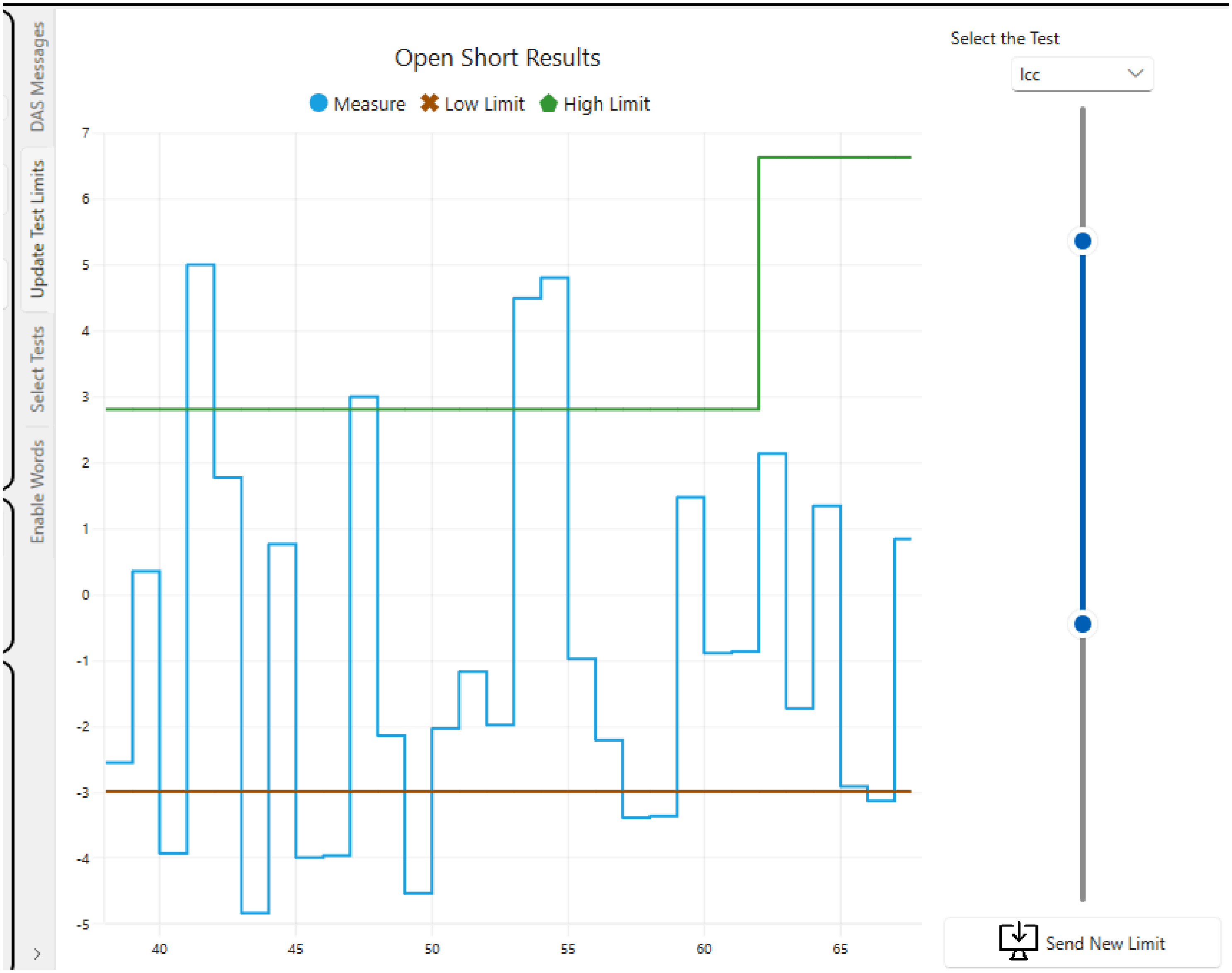
DAS Section

View all messages generated by the tester. Clicking on a message reveals its detailed contents.



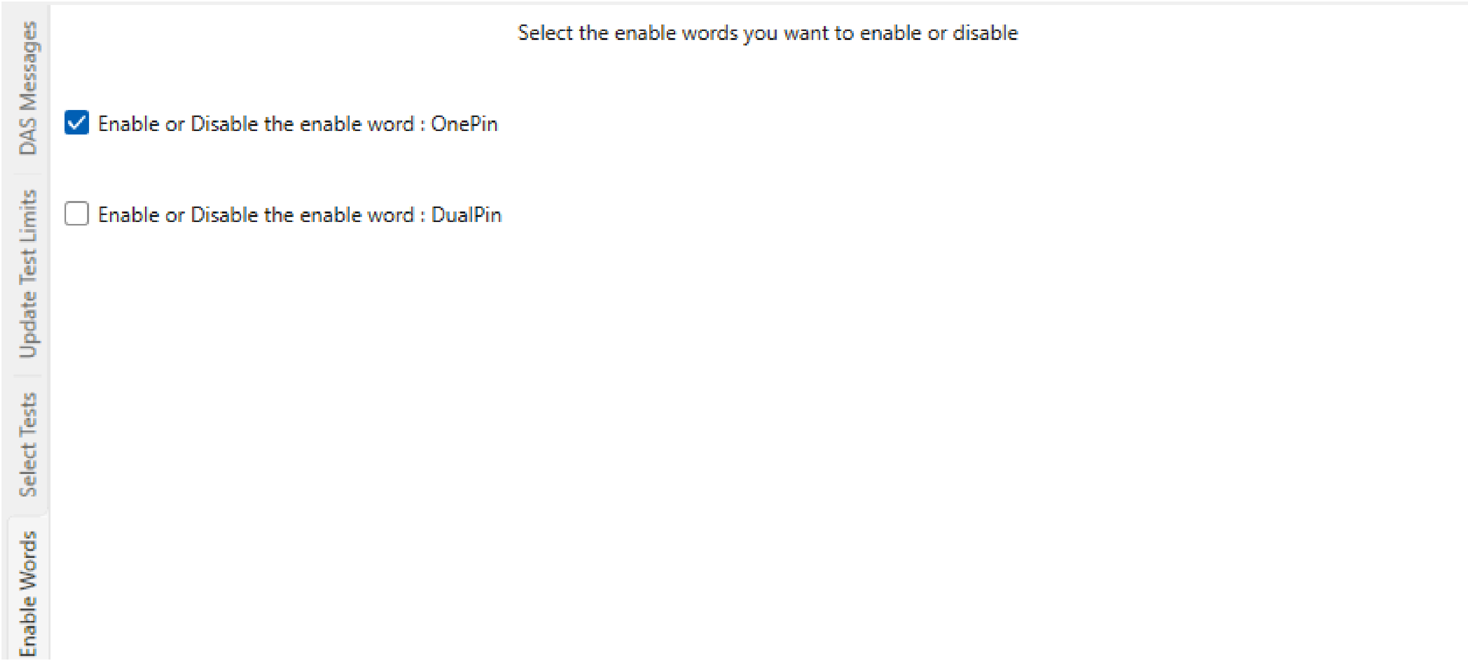
Test Limits Section

Displays the test results for `Icc` and its associated limits. Use the slider and **Update Test Limits** button to adjust the thresholds.



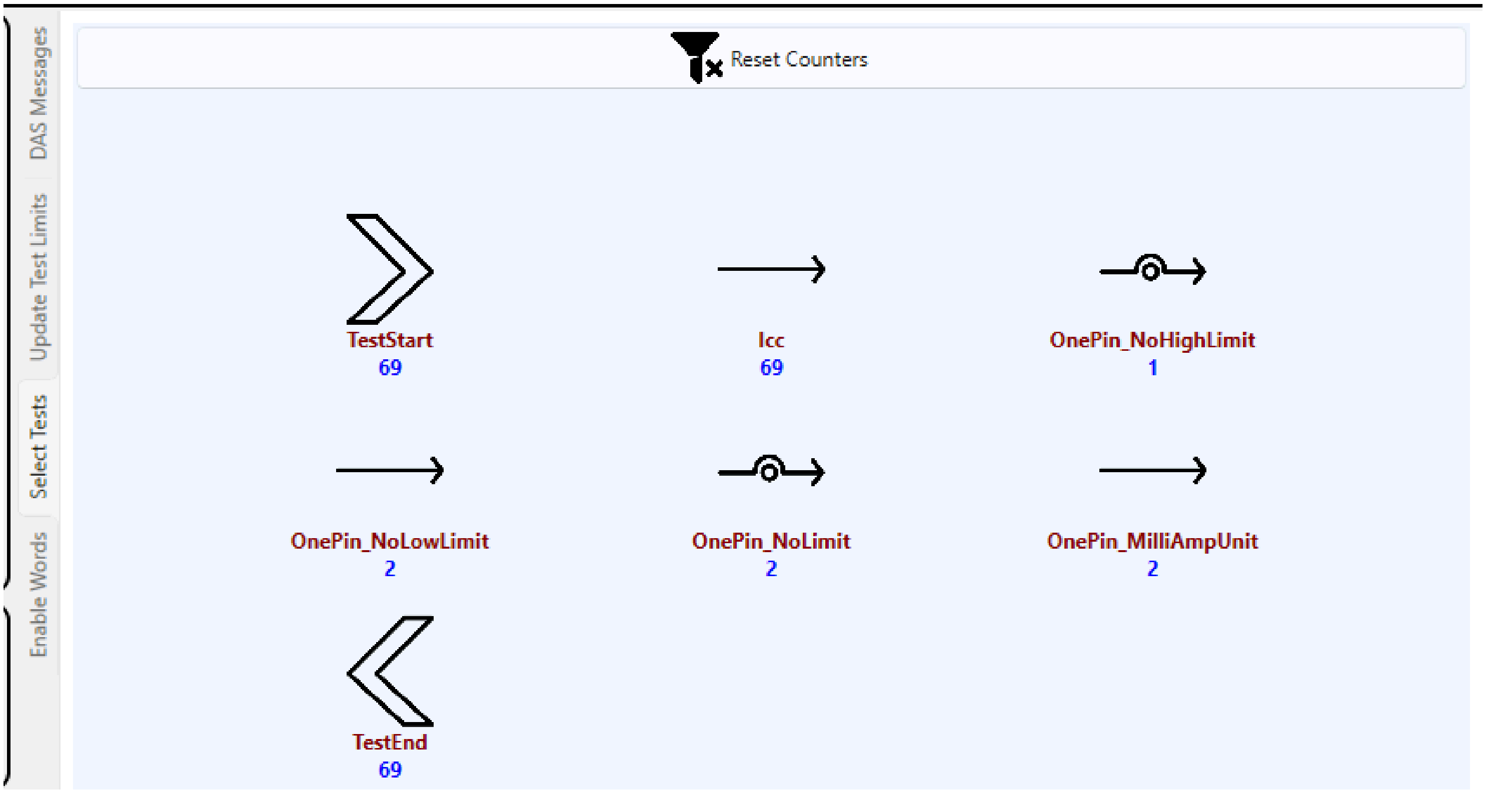
Enable Word Section

The test program includes two enable words that control sub-flow activation. Enable or disable them directly from this interface.



Test Enable/Disable Section

Enable or disable tests dynamically with a simple click in the interface.



Technical Snapshot

Component	Version
Language	C#

Component	Version
Framework	.NET 8
Environment	Windows 10
Excel	2016 (64-bit)
IGXL	10.60.02

Extend the Demo to Your Environment

To implement a similar system, we recommend reviewing the following guides:

Reference Architecture	Description
RA_0468	How to implement a DAS in C#
RA_0478	How to send an adaptive command in C#

See It in Action

Explore the complete Instructions to Run the Demo or browse the image gallery to discover key functionalities.

Looking to integrate this approach into your test strategy? Contact our team to explore how DAS can support your roadmap.

This demo is part of our Reference Architecture Series – built to inspire smarter test development.

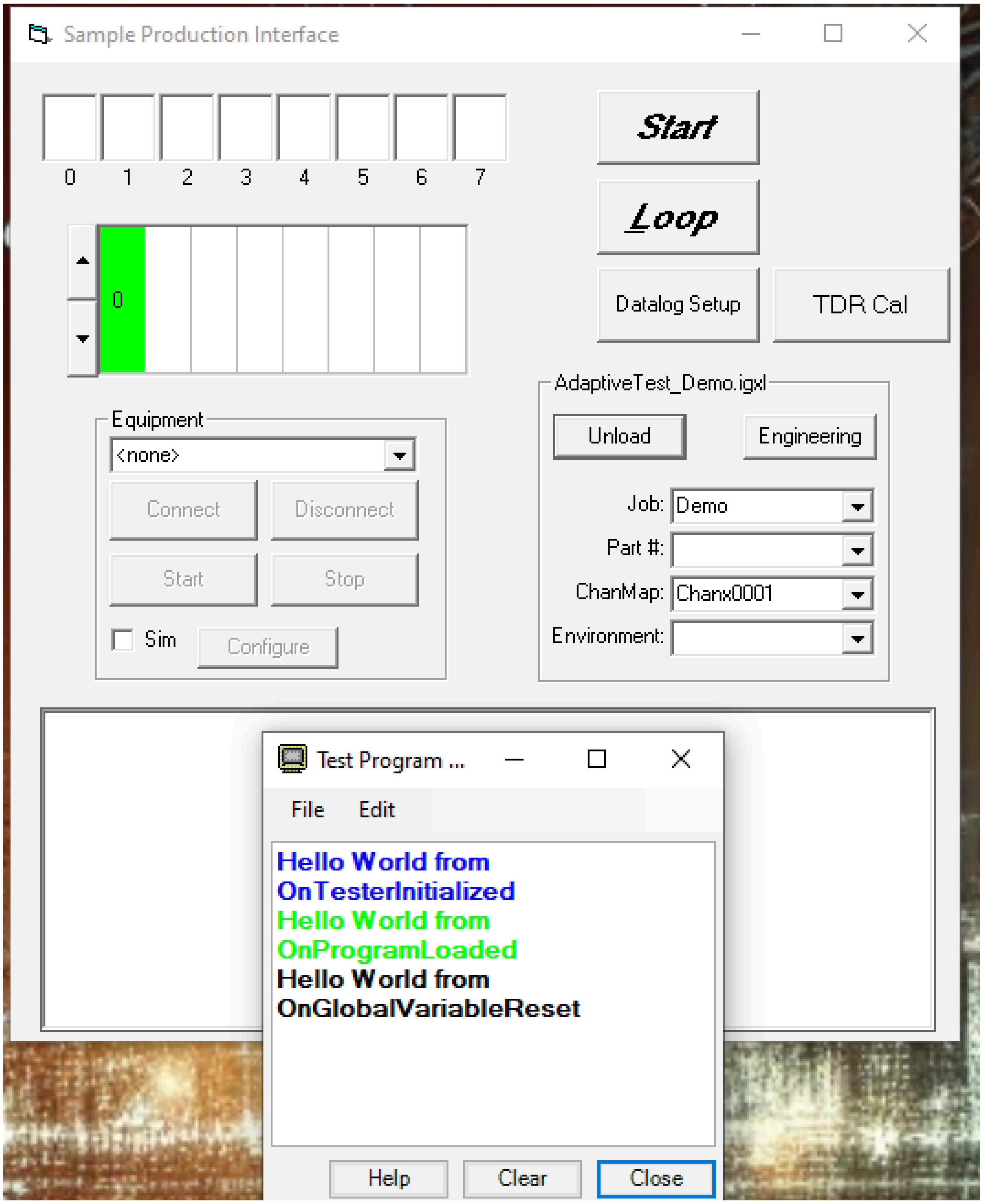
How to Run the Adaptive Test Demo

Reference: RA_453

This section provides step-by-step instructions to explore the key features of the Adaptive Test DAS Demo.

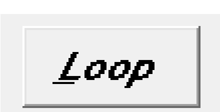
Step 1: Load the Test Program

Use the SimpleOI interface to load your test program.



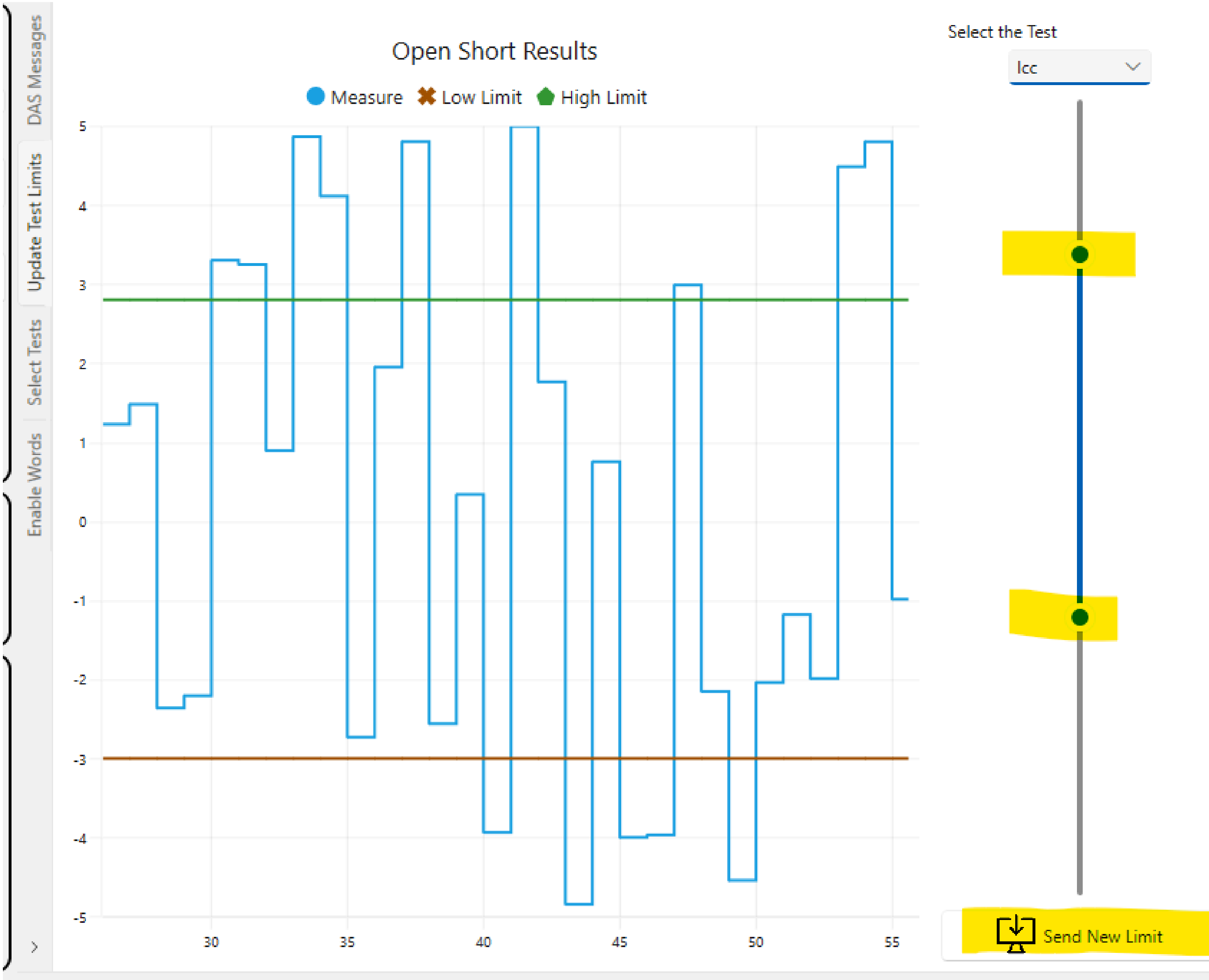
Step 2: Execute the Program in Loop Mode

Click the **Loop** button to execute the program 10 times. Once completed, press the **Stop** button.



Step 3: Adjust Test Limits Dynamically

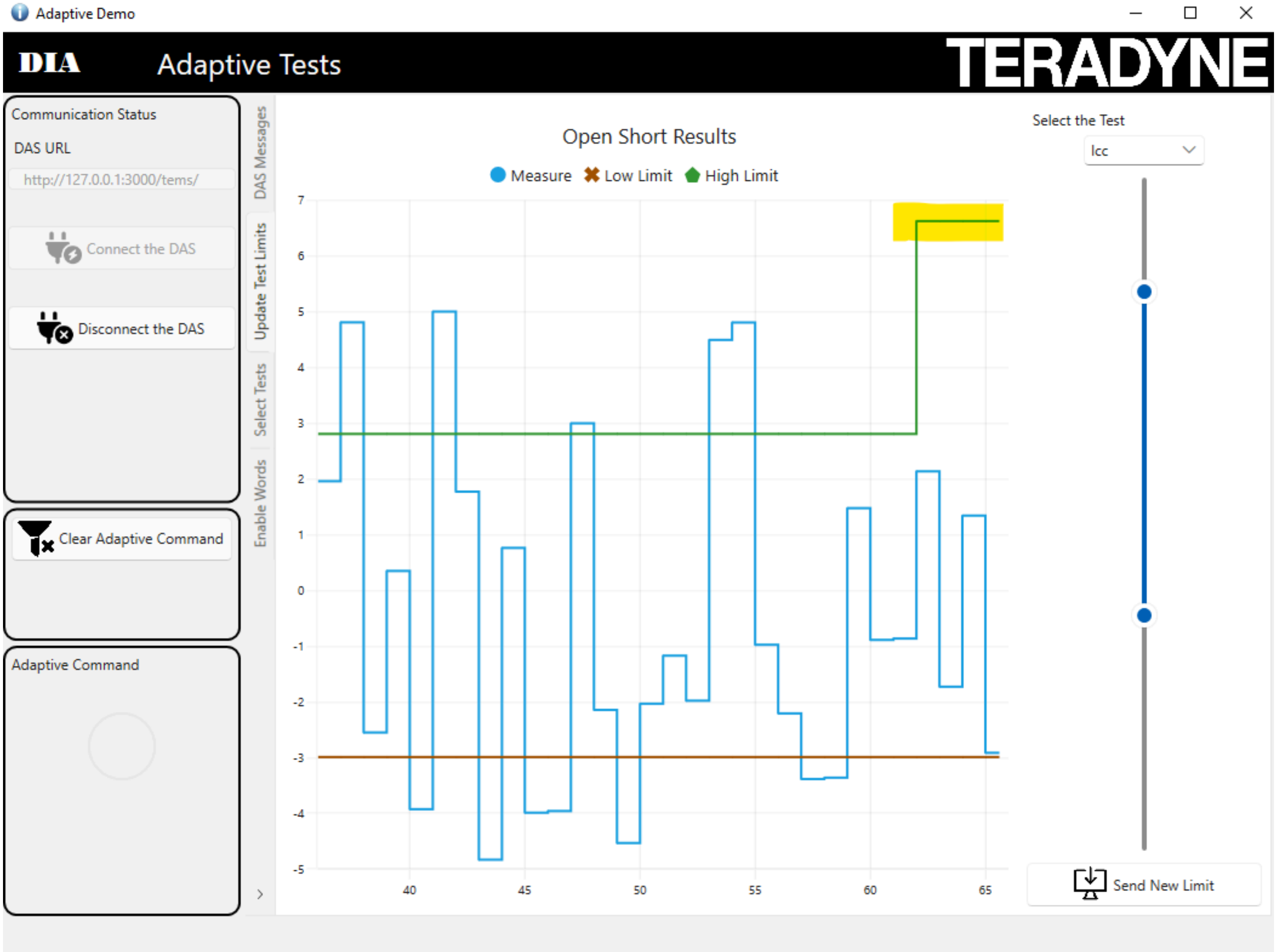
In the demo application, click **Update Test Limits**. Use the slider to modify the thresholds, then click **Send Limits** to apply them.



Run the test a few times from the SimpleOI interface to observe the impact.

Step 4: Observe the Updated Limits

The graphical output will reflect the new limit values.



In DAS, select the `Adaptive_Change limits` message. Click on the message to inspect its contents.

Time Stamp	Test Cell	Name	Content
08:01:48.9137	TEMS-01	ADAPTIVE_CHANGE	TEP_ACTIONS": [{"ACTION":"UPDATE_LIMITS","REQUEST_ID":1,"TESTSTEP_NAMES": ["Icc"],"LIMITS":{"LIMIT_NAMES":


```
{
 "TIME_STAMP": "2025-05-05T08:01:48.9137009Z",
 "LOT_ID": "UNKNOWNLOT_PID_8380_1",
 "SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
 "DAS_ID": 1,
 "TESTSTEP_ACTIONS": [
  {
   "ACTION": "UPDATE_LIMITS",
   "REQUEST_ID": 1,
   "TESTSTEP_NAMES": [
    "Icc"
   ],
   "LIMITS": {
    "LIMIT_NAMES": [
     "Icc"
```


You can also observe the received message in the IGXL test program output.

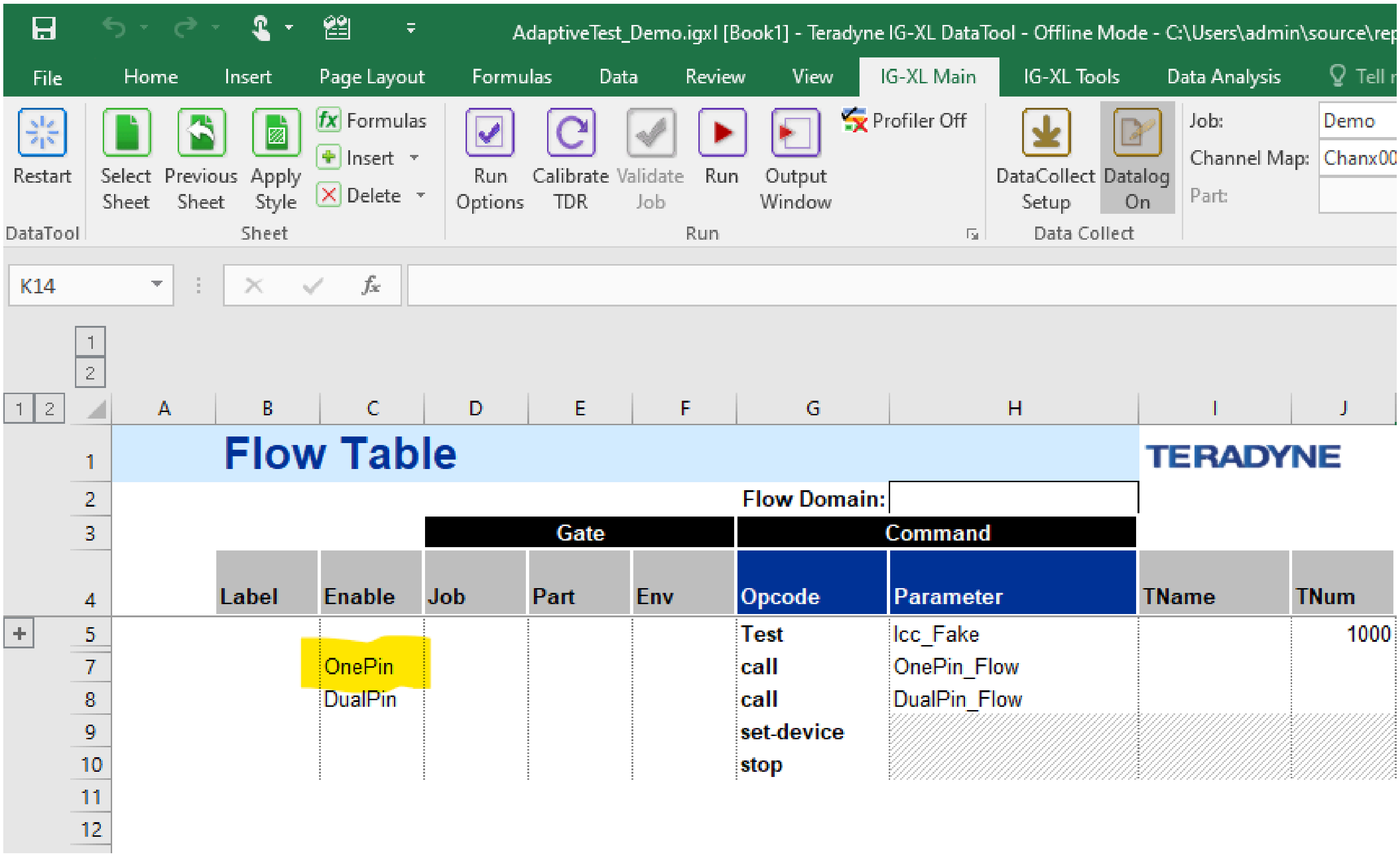
```
Teradyne_IG-XL_DataCollect_8380 - OutputWin
File  Edit  Collect  View  Help

 Site    Sort     Bin
------------------------------------
    0       1       1
==========================================================================
OnProgramEnded: 2025-05-05 10:01:40.390
Adaptive Change: The limits {Icc} on test step {Icc} were changed. New low limit = -3. New high limit = 6.61954453733064.

OnProgramStarted: 2025-05-05 10:01:48.910
    Device#: 63

 Number   Site  Test Name      Pin  Channel  Low            Measured       High           Force         Loc
<Icc_Fake>
 1000     0     Icc                 -1       -3000.0000 mA  2127.3046 mA   6619.5445 mA   0.0000        0
```

Step 5: Work with Enable Words

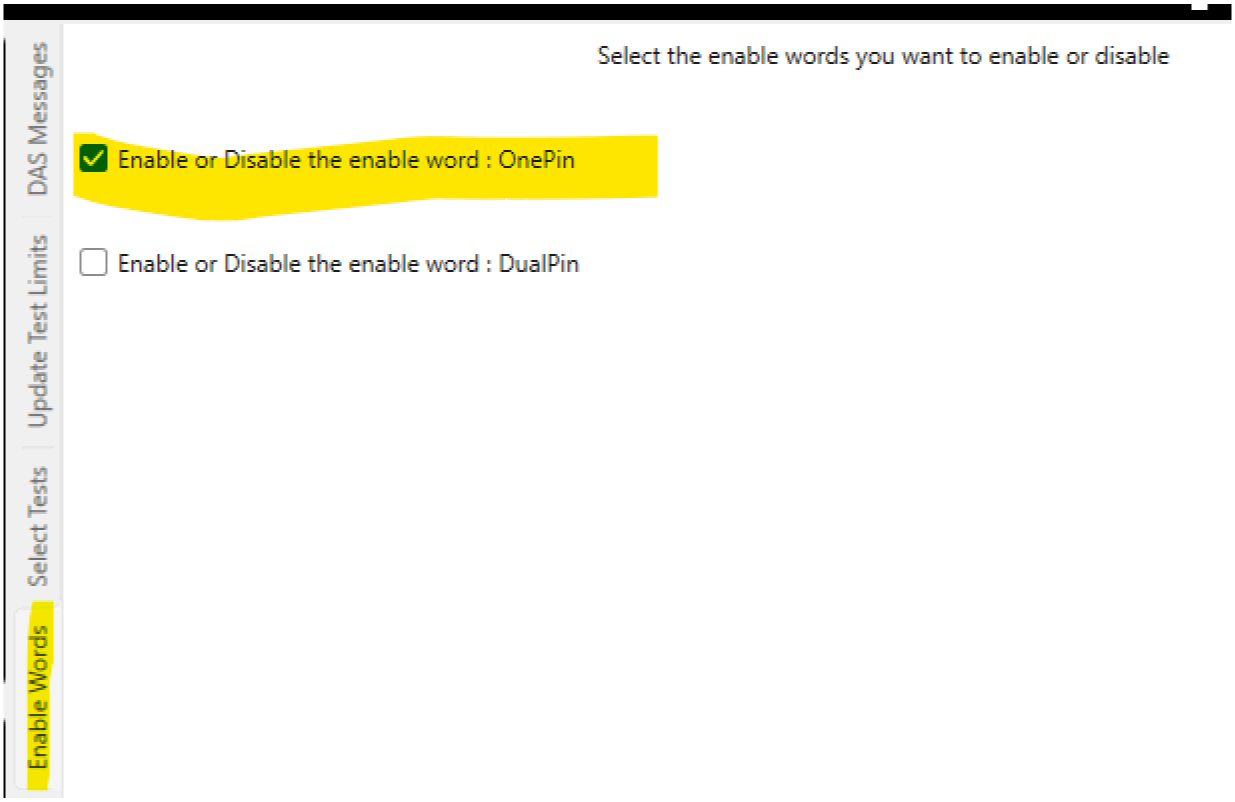
This part demonstrates how to activate a sub-flow using enable words. Initially, only the test `ICC` is active.

To explore more:

- Access the engineering view in SimpleOI (password: `Teradyne`)
- Open the test program and locate the `DemoFlow`
- Visualize the flow and locate the enable word logic



Click on the `OnePin EnableWord` option and run the test program once.



You'll see that more tests are now activated:

```
Adaptive Change: The following enable words were enabled: {OnePin}.

OnProgramStarted: 2025-05-05 10:09:38.800
    Device#: 68


 Number   Site  Test Name       Pin  Channel  Low            Measured          High          Force         Loc
<Icc_Fake>
 1000      0     Icc                  -1       -3000.0000 mA  833.5900 mA       6619.5445 mA  0.0000        0
OnePin
<OnePin_NoHighLimit>
 200       0     OnePin_NoHighLi      -1       100.0000 m     1.1235            N/A           0.0000        0
<OnePin_NoLowLimit>
 300       0     OnePin_NoLowLim      -1       N/A            500.0000 m        900.0000 m    0.0000        0
<OnePin_NoLimit>
 400       0     OnePin_NoLimit       -1       N/A            -700.0000 m       N/A           0.0000        0
<OnePin_MilliAmpUnit>
 500       0     OnePin_MilliAmp      -1       -1000000000.0000 nA 17500000000.0000 nA (F) 0.0000 nA      0.0000         0
```

A new Adaptive Change message will also be generated:

08:09:38.8061	TEMS-01	ADAPTIVE_CHANGE	ID_8380_1","SUBLOT_ID":"UNKNOWNSUBLOT_PID_8380_1","DAS_ID":1,"TESTSTEP_ACTIONS":[],"ENABLEWORD_ACTIONS":[{"ACTION":"ENABLE","REQUEST_ID":2,"ENABLEWORDS":

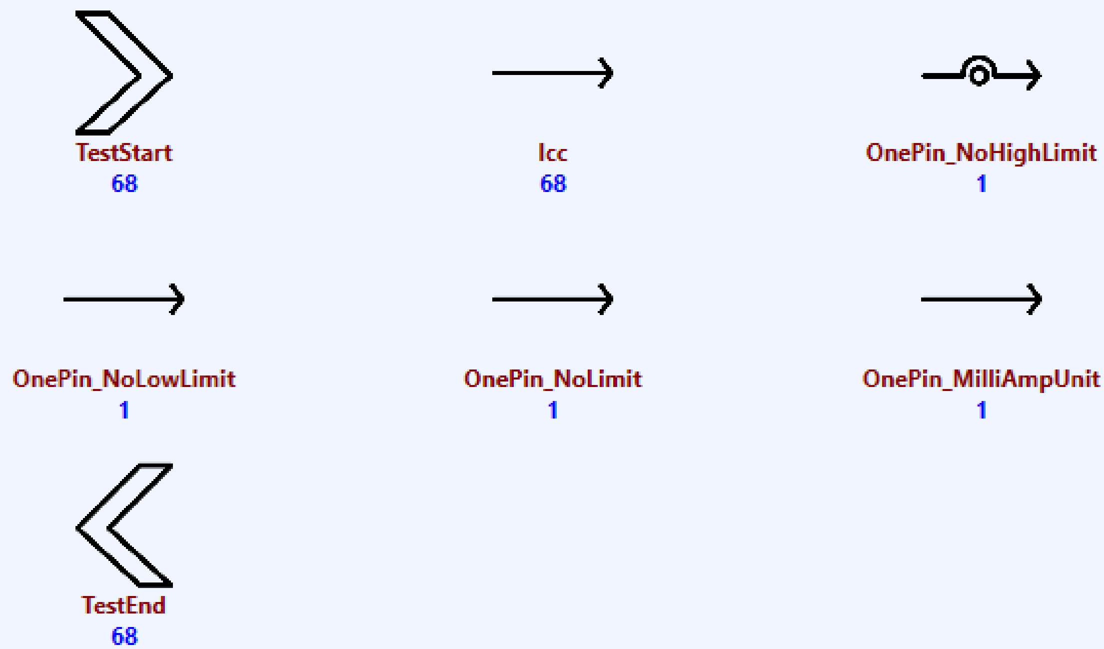

```
"TIME_STAMP": "2025-05-05T08:09:38.806154Z",
"LOT_ID": "UNKNOWNLOT_PID_8380_1",
"SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
"DAS_ID": 1,
"TESTSTEP_ACTIONS": [],
"ENABLEWORD_ACTIONS": [
 {
   "ACTION": "ENABLE",
   "REQUEST_ID": 2,
   "ENABLEWORDS": [
     "OnePin"
   ],
   "STATUS": "OK"
 }
```


Step 6: Enable or Disable Tests

This step shows how individual tests can be selectively enabled or disabled.

- Go to the **Select Test** tab in the DAS interface
- A count of executions for each test is shown in blue
- For best demo results, activate **DoAll** in IGXL Run Options

Now, disable the `OnePin_NoHighLimit` test by selecting it.



Run the test program once. You'll see that the test is skipped.

08:25:20.0344	TEMS-01	ADAPTIVE_CHANGE	ID_8380_1","SUBLOT_ID":"UNKNOWNSUBLOT_PID_8380_1","DAS_ID":1,"TEST TEP_ACTIONS":[{"ACTION":"DISABLE","REQUEST_ID":3,"TESTSTEP_NAMES":

```
{
  "TIME_STAMP": "2025-05-05T08:25:20.034446Z",
  "LOT_ID": "UNKNOWNLOT_PID_8380_1",
  "SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
  "DAS_ID": 1,
  "TESTSTEP_ACTIONS": [
   {
    "ACTION": "DISABLE",
    "REQUEST_ID": 3,
    "TESTSTEP_NAMES": [
     "OnePin_NoHighLimit"
    ],
    "STATUS": "OK"
   }
  ],
```

You can confirm this in IGXL:

```
OnProgramEnded: 2025-05-05 10:09:38.870
Adaptive Change: The following test steps were disabled: {OnePin_NoHighLimit}.

OnProgramStarted: 2025-05-05 10:25:20.030
    Device#: 69

 Number    Site   Test Name        Pin  Channel  Low             Measured            High           Force          Loc
<Icc_Fake>
 1000      0      Icc                   -1       -3000.0000 mA  -4192.8535 mA  (F) 6619.5445 mA   0.0000         0
OnePin
<OnePin_NoLowLimit>
 300       0      OnePin_NoLowLim       -1       N/A             500.0000 m          900.0000 m     0.0000         0
<OnePin_NoLimit>
 400       0      OnePin_NoLimit        -1       N/A             -700.0000 m         N/A            0.0000         0
<OnePin_MilliAmpUnit>
 500       0      OnePin_MilliAmp       -1       -1000000000.0000 nA 17500000000.0000 nA (F) 0.0000 nA      0.0000         0
```

Download the Demo

Download the demo from here